THE STACK OF CURVES IS OPEN AND CLOSED IN THE STACK OF PROPER SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equation quot-equation-curves-over-proper-spaces (canonical tag 0D50). cited at statement line 5037. The fully faithful embedding.
- Lemma spaces-more-morphisms-lemma-dimension-fibres-proper-flat (canonical tag 0D4R). cited at proof line 5053. Dimension of fibres of proper flat morphisms.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at statement line 5037: the fully faithful embedding.

Source reference. Original lemma 0D4R: sources/stacks-chow/tags.txt, source lines 16461–16461. Cited at statement line 5053: relative dimension <= 1 locus is open and closed.

Lemma 0.2. *The* 1*-morphism (0D50)*

$$\mathcal{C}urves \longrightarrow \mathcal{S}paces'_{fp,flat,proper}$$

is representable by open and closed immersions.

Proof. Since (0D50) is a fully faithful embedding of categories it suffices to show the following: given an object $X \to S$ of $\mathcal{S}paces'_{fp,flat,proper}$ there exists an open and closed subscheme $U \subset S$ such that a morphism $S' \to S$ factors through U if and only if the base change $X' \to S'$ of $X \to S$ has relative dimension ≤ 1. This follows immediately from More on Morphisms of Spaces, Lemma 0D4R. □

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